

$$\begin{aligned} & e_{\phi_1}^{11} \Gamma_{-12} \rightarrow \frac{1}{16\pi^2} \left(-\frac{1}{3} m_2 \tilde{\mu} g_2^4 y_e^{11i2} \text{LF}_{2,2,0}[m_2, \tilde{\mu}] + \right. \\ & \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_e^{11i2} \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \frac{1}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} y_e^{11i2} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,0}[m_d^r, m_q^{\text{p}}] + \\ & \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} y_e^{11i2} \overline{a_e^{\text{pr}}} \text{LF}_{2,2,0}[m_e^r, m_l^{\text{p}}] - \frac{1}{2} \tilde{\mu} g_1^2 y_e^{\text{pr}} y_e^{11i2} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0}[m_l^{\text{p}}, m_e^r] - \\ & \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} y_e^{11i2} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1}[m_l^{\text{p}}, m_e^r] + \frac{3}{8} \tilde{\mu} y_e^{\text{pr}} y_e^{11i2} \overline{a_e^{\text{pr}}} (3 g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_l^{\text{p}}, m_e^r] - \\ & \frac{1}{6} \tilde{\mu} y_e^{\text{pr}} y_e^{11i2} \overline{a_e^{\text{pr}}} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2}[m_l^{\text{p}}, m_e^r] - \\ & \frac{1}{2} \tilde{\mu} g_1^2 y_d^{\text{pr}} y_e^{11i2} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0}[m_q^{\text{p}}, m_d^r] - \frac{1}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} y_e^{11i2} \overline{a_d^{\text{pr}}} \text{LF}_{3,2,-1}[m_q^{\text{p}}, m_d^r] + \\ & \frac{3}{8} \tilde{\mu} y_d^{\text{pr}} y_e^{11i2} \overline{a_d^{\text{pr}}} (5 g_1^2 + 3 g_2^2) \text{LF}_{4,1,-1}[m_q^{\text{p}}, m_d^r] - \\ & \frac{1}{2} \tilde{\mu} y_d^{\text{pr}} y_e^{11i2} \overline{a_d^{\text{pr}}} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2}[m_q^{\text{p}}, m_d^r] + \\ & \frac{1}{8} \tilde{\mu} y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 - 5 g_2^2) \text{LF}_{2,2,0}[m_q^{\text{p}}, m_u^r] - \frac{3}{4} \tilde{\mu} g_2^2 y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0}[m_q^{\text{p}}, m_u^r] + \\ & \frac{1}{2} \tilde{\mu} g_2^2 y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,2,-1}[m_q^{\text{p}}, m_u^r] + \frac{3}{4} \tilde{\mu} g_2^2 y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{4,1,-1}[m_q^{\text{p}}, m_u^r] - \\ & \frac{1}{4} \tilde{\mu} y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (7 g_1^2 + 2 g_2^2) \text{LF}_{3,1,0}[m_u^r, m_q^{\text{p}}] - \\ & \frac{1}{8} \tilde{\mu} y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 - 5 g_2^2) \text{LF}_{3,2,-1}[m_u^r, m_q^{\text{p}}] + \\ & \frac{3}{4} \tilde{\mu} y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (5 g_1^2 + 2 g_2^2) \text{LF}_{4,1,-1}[m_u^r, m_q^{\text{p}}] - \frac{1}{2} \tilde{\mu} y_e^{11i2} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \\ & (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2}[m_u^r, m_q^{\text{p}}] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (g_1^2 + g_2^2) \text{LF}_{3,1,0}[\tilde{\mu}, m_1] + \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{11i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (6 g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{11i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] - \\ & \frac{1}{12} m_2 \tilde{\mu} g_2^2 y_e^{11i2} (9 g_1^2 + 5 g_2^2) \text{LF}_{3,1,0}[\tilde{\mu}, m_2] + \frac{7}{3} m_2 \tilde{\mu} g_2^4 y_e^{11i2} \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^2 y_e^{11i2} (18 g_1^2 + 11 g_2^2) \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] - \\ & 2 m_2 \tilde{\mu} g_2^4 y_e^{11i2} \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] - \frac{1}{2} m_2 \tilde{\mu} g_2^2 y_e^{11i2} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] + \\ & \frac{1}{16} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_1, m_e^{i2}, m_l^{i1}] + \\ & \frac{1}{16} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_1, m_l^{i1}, m_e^{i2}] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (-7 g_1^2 + g_2^2) \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i2}] + \\ & \frac{1}{16} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (7 g_1^2 - g_2^2) \text{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{i1}] + \\ & \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 y_e^{11i2} (7 m_1 + 5 m_2) \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\ & \tilde{\mu} g_1^2 g_2^2 y_e^{11i2} (m_1 + m_2) \text{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{3}{16} m_2 \tilde{\mu} g_2^2 y_e^{11i2} (g_1^2 - 7 g_2^2) \\ & \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i1}] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (3 g_1^2 - g_2^2) \text{LF}_{2,1,1,0}[m_e^{i2}, m_1, m_l^{i1}] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_e^{i2}, m_1, m_l^{i1}] - \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_e^{11i2} \text{LF}_{2,1,1,0}[m_e^{i2}, m_1, \tilde{\mu}] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (-3 g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_l^{i1}, m_1, m_e^{i2}] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1}[m_l^{i1}, m_1, m_e^{i2}] + \\ & \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_e^{11i2} (-g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_l^{i1}, m_1, \tilde{\mu}] + \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^2 y_e^{11i2} (3 g_1^2 + g_2^2) \text{LF}_{2,1,1,0}[m_l^{i1}, m_2, \tilde{\mu}] - \\ & \frac{3}{4} \tilde{\mu} y_d^{\text{sr}} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_d^r] + \\ & \frac{3}{4} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^r] - \\ & \frac{3}{4} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_q^{\text{p}}, m_q^{\text{s}}, m_u^r] - \\ & \frac{9}{8} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{p}}, m_u^r, m_q^{\text{s}}] + \\ & \frac{3}{4} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0}[m_q^{\text{s}}, m_q^{\text{p}}, m_u^r] - \\ & \frac{3}{4} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,1,-1}[m_q^{\text{s}}, m_q^{\text{p}}, m_u^r] + \\ & \frac{3}{8} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1}[m_q^{\text{s}}, m_u^r, m_q^{\text{p}}] + \\ & \frac{3}{2} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0}[m_u^r, m_q^{\text{p}}, m_q^{\text{s}}] - \\ & \frac{3}{2} \tilde{\mu} y_e^{11i2} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}}$$